



Yakeen Plus (2025)

SHORT PRACTICE TEST - 04

DURATION: 60 Minutes

DATE : 06/10/2024

M.MARKS : 192

Topics Covered

Physics:	Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluids
Chemistry:	Electrochemistry Chemical Kinetics Structure of Atom
Biology:	(Botany) : Biological Classification (Zoology) : Excretory Products & their Elimination

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **60 minutes** duration.
3. The test booklet consists of **48** questions. The maximum marks are **192**.
4. All questions are compulsory.
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

SECTION-(I) PHYSICS

1. If the distance between two masses is doubled, the gravitational attraction between them:

(1) is doubled.
 (2) becomes four times.
 (3) is reduced to half.
 (4) is reduced to a quarter.

2. If the change in the value of 'g' at a height h above the surface of the earth is the same as at a depth x below it, then (both x and h being much smaller than the radius of the earth):

(1) $x = h$ (2) $x = 2h$
 (3) $x = \frac{h}{2}$ (4) $x = h^2$

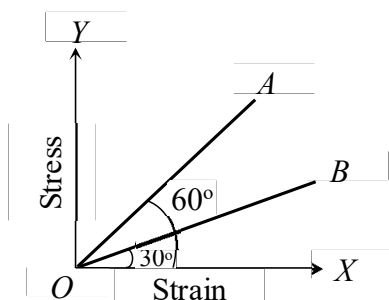
3. For a satellite escape velocity is 11 km/s. If the satellite is launched at an angle of 60° with the vertical, then escape velocity will be:

(1) 11 km/s (2) $11\sqrt{3}$ km/s
 (3) $\frac{11}{\sqrt{3}}$ km/s (4) 33 km/s

4. The ratio of the lengths of two wires A and B of same material is 1 : 2 and the ratio of their diameter is 2 : 1. They are stretched by the same force, then the ratio of increase in length will be:

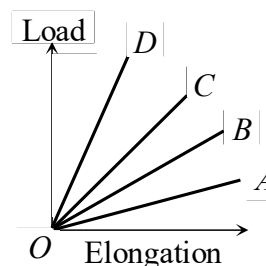
(1) 2 : 1 (2) 1 : 4
 (3) 1 : 8 (4) 8 : 1

5. The stress versus strain graphs for wires of two materials A and B are as shown in the figure. If Y_A and Y_B are the Young's modulus of the materials, then:



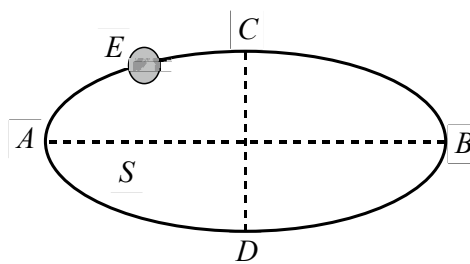
(1) $Y_B = 2Y_A$
 (2) $Y_A = Y_B$
 (3) $Y_B = 3Y_A$
 (4) $Y_A = 3Y_B$

6. The load versus elongation graph for four wires of the same material and same lengths is shown in the figure. The thickest wire is represented by the line:



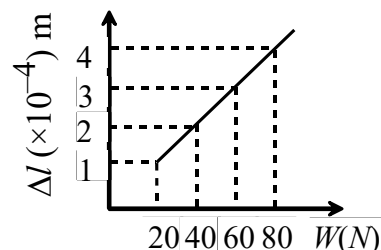
(1) OD (2) OC
 (3) OB (4) OA

7. The earth E moves in an elliptical orbit with the sun S at one of the foci as shown in figure. Its speed of motion will be maximum at the point:



(1) C (2) A
 (3) B (4) D

8. The adjacent graph shows the extension (Δl) of a wire of length 1m suspended from the top of a roof at one end with a load W connected to the other end. If the cross-sectional area of the wire is 10^{-6} m^2 , calculate the young's modulus of the material of the wire:

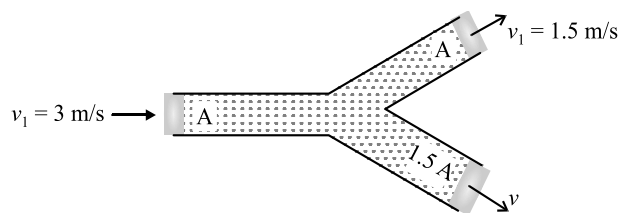


(1) $2 \times 10^{11} \text{ N/m}^2$ (2) $2 \times 10^{-11} \text{ N/m}^2$
 (3) $3 \times 10^{-12} \text{ N/m}^2$ (4) $3 \times 10^{-13} \text{ N/m}^2$

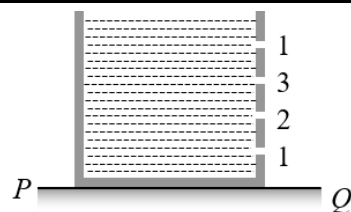
9. If pressure at half the depth of a lake is equal to $2/3$ pressure at the bottom of the lake then what is the depth of the lake:

(1) 10 m
 (2) 20 m
 (3) 60 m
 (4) 30 m

10. An incompressible liquid flows through a horizontal tube as shown in the following fig. Then the velocity v of the fluid is:

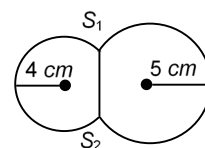


- (1) 3.0 m/s
(2) 1.5 m/s
(3) 1.0 m/s
(4) 2.25 m/s
11. A cylindrical vessel of 90 cm height is kept filled upto the brim. It has four holes 1, 2, 3, 4 which are respectively at heights of 20 cm, 30 cm, 45 cm and 50 cm from the horizontal floor PQ. The water falling at the maximum horizontal distance from the vessel comes from:



- (1) Hole number 4
(2) Hole number 3
(3) Hole number 2
(4) Hole number 1

12. Two soap bubbles of radii r_1 and r_2 equal to 4 cm and 5 cm are touching each other over a common surface S_1S_2 (shown in figure). Its radius will be:



- (1) 4 cm
(2) 20 cm
(3) 5 cm
(4) 4.5 cm

[Short Practice Test-04 | Yakeen Plus (2025) | 06/10/2024]
Electrochemistry Chemical Kinetics Structure of Atom

SECTION-(II) CHEMISTRY

13. If a half-cell X/X^{2+} (0.1 M) is connected to another half-cell Y/Y^{2+} (1.0 M) by means of a salt bridge and an external circuit at 25°C , the cell voltage would be:

(Given: $E^\circ_{Y^{2+}/Y} = 0.34\text{V}$, $E^\circ_{X^{2+}/X} = -0.25\text{V}$)

- (1) 0.26 V
(2) 0.31 V
(3) 0.62 V
(4) 0.72 V
14. For the cell:
 $\text{Ni} | \text{Ni}^{2+} || \text{Cu}^{2+} | \text{Cu}$; $E^\circ = 0.77\text{ V}$.
By which of the following activity, E_{cell} will increase?
- (1) On decreasing $[\text{Ni}^{2+}]$
(2) On decreasing $[\text{Cu}^{2+}]$
(3) On increasing mass of Ni electrode
(4) On increasing mass of Cu electrode
15. The compound exhibiting maximum value of equivalent conductance in fused state is:
- (1) SrCl_2
(2) CaCl_2
(3) MgCl_2
(4) BeCl_2
16. What is the quantity of electricity (in coulombs) required to deposit all the silver from 250 mL of 1 M AgNO_3 solution?
(Atomic mass of Ag = 108)
- (1) 2412.5
(2) 24125
(3) 4825.0
(4) 28250

17. The integrated rate equation is $\text{Rt} = \log C_0 - \log C_t$. The straight line graph is obtained by plotting.

- (1) time vs. $\log C_t$
(2) $\frac{1}{\text{time}}$ vs. C_t
(3) time vs. C_t
(4) $\frac{1}{\text{time}}$ vs. $\frac{1}{C_t}$

18. The rate constant k_1 and k_2 for two different reaction are $10^{16} \cdot e^{-2000/T}$ and $10^{15} \cdot e^{-1000/T}$, respectively. The temperature at which $k_1 = k_2$ is:

- (1) 1000 K
(2) $\frac{2000}{2.303}\text{K}$
(3) 2000 K
(4) $\frac{1000}{2.303}\text{K}$

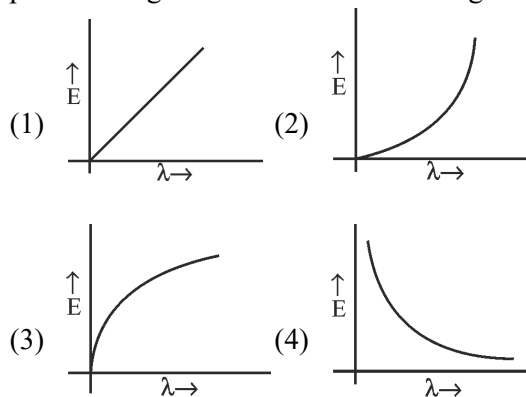
19. $2\text{A} + \text{B} \rightarrow \text{C} + \text{D}$

In this reaction, if we double the concentration of A, reaction rate become two times. And in other experiment, we double the concentration of A and B, reaction rate again become two times. What is the order of this reaction?

- (1) 1
(2) 3
(3) 2
(4) 1.5

20. For an endothermic reaction where ΔH represent the enthalpy of reaction in kJ mol^{-1} , the activation energy of forward reaction will be:
- less than ΔH
 - zero
 - more than ΔH
 - equal to ΔH
21. Bohr's model may be applied to:
- Na^{10+} ion
 - He atom
 - Be^{2+} ion
 - C^{6+} ion
22. The ratio of minimum to maximum wavelengths in Balmer series of H-atom is:
- 5 : 9
 - 5 : 36
 - 1 : 4
 - 3 : 4

23. Which graph shows how the energy E of a photon of light is related to its wavelengths (λ)?



24. The probability of finding an electron in an orbital is approximately:
- 95 %
 - 5 %
 - 6 %
 - 25 %

[Short Practice Test-04 | Yakeen Plus (2025) | 06/10/2024]
Biological Classification

SECTION-(III) BOTANY

25. Mycoplasma are organisms that:
- lack a cell wall and cannot survive without oxygen.
 - lack a cell membrane and can survive without oxygen.
 - lack a cell wall and can survive without oxygen.
 - lack a cell membrane and cannot survive without oxygen.
26. Select the **correct** statement/s w.r.t protozoans.
- All are heterotrophs.
 - Live as predators or parasites.
 - Included in kingdom Monera.
 - Primitive relatives of animals.
- Choose the most appropriate answer from the options given below.
- A and B only.
 - B and C only.
 - A, B and D only.
 - C and D only.
27. Which of the following is **not** true for diatoms?
- Float passively in water currents.
 - They are large sized chrysophytes.
 - Cell walls are indestructible.
 - Used in polishing, filtration of oils and syrups.
28. Given below are two statements:
- Statement I:** Organisms belongs to kingdom animalia follow a definite growth pattern.
- Statement II:** Higher forms of animalia kingdom shows elaborate sensory and neuromotor mechanism.

In the light of the above statements, choose the most appropriate answer from the options given below.

- Statement I is correct but Statement II is incorrect.
 - Statement I is incorrect but Statement II is correct.
 - Both Statement I and Statement II are correct.
 - Both Statement I and Statement II are incorrect.
29. Mark the **incorrectly** matched pair.
- Phycomycetes - Aseptate hyphae
 - Ascomycetes - Septate hyphae
 - Basidiomycetes - Unbranched mycelium
 - Deuteromycetes - Branched mycelium
30. Match **List-I** with **List-II**.
- | List I | | List II | |
|--------|-----------------|---------|----------------------------|
| (A) | <i>Ustilago</i> | (I) | Mushroom |
| (B) | <i>Puccinia</i> | (II) | Parasitic fungi on mustard |
| (C) | <i>Agaricus</i> | (III) | Smut |
| (D) | <i>Albugo</i> | (IV) | Rust |
- Choose the **correct** answer from the options given below.
- A-II, B-IV, C-III, D-I
 - A-I, B-II, C-III, D-IV
 - A-II, B-I, C-IV, D-III
 - A-III, B-IV, C-I, D-II

31. Select the **correctly** matched pair.
- (1) Photosynthetic – Diatoms and slime moulds
 - (2) Marine – Diatoms and euglenoids
 - (3) Flagellates – Diatoms and dinoflagellates
 - (4) Saprophytic – Slime moulds and fungi
32. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: In slime moulds, plasmodium shows differentiation in favourable conditions.
Reason R: Spores of slime mould have true walls.
 In the light of the above statements, choose the **correct** answer from the options given below.
- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true but R is NOT the correct explanation of A.
33. In basidiomycetes, karyogamy and meiosis take place in the basidium producing:
- (1) two basidiospores.
 - (2) one basidiospore.
 - (3) many basidiospores.
 - (4) four basidiospores.

34. Match **List-I** with **List-II**.

List I		List II	
(A)	Viroids	(I)	Mad cow disease in cattle
(B)	Bacteriophage	(II)	Single stranded RNA virus
(C)	Prions	(III)	Potato spindle tuber disease
(D)	Tobacco Mosaic Virus	(IV)	Double stranded DNA

Choose the **correct** answer from the options given below.

- (1) A-II, B-IV, C-III, D-I
 - (2) A-I, B-II, C-III, D-IV
 - (3) A-III, B-IV, C-I, D-II
 - (4) A-III, B-IV, C-II, D-I
35. Select the **correct** statements. Tobacco Mosaic Virus
- A. Bacteria adopt sexual reproduction by gametic fusion only.
 - B. In ascomycetes, sexual spores are known as ascospores.
 - C. Sac-fungi are commonly known forms of basidiomycetes.
 - D. Group of fungi which produces endogenous asexual spores is phycomycetes.
 - E. Under favourable conditions, the most common method of reproduction in bacteria is fission.
- Choose the most appropriate answer from the options given below.
- (1) A and B only.
 - (2) B and C only.
 - (3) A, B, D and E only.
 - (4) B, D and E only.
36. An infectious spore - like stage is present in the life cycle of a protozoan which causes:
- (1) sleeping sickness.
 - (2) AIDS.
 - (3) encephalopathy.
 - (4) malaria.

[Short Practice Test-04 | Yakeen Plus (2025) | 06/10/2024]
 Excretory Products & their Elimination

SECTION-(IV) ZOOLOGY

37. Urine is stored in the urinary bladder until which of the following triggers its release?
- (1) A voluntary signal from the CNS
 - (2) An involuntary signal from the CNS
 - (3) A voluntary signal from the ANS
 - (4) An involuntary signal from the ANS
38. Given below are two statements:
- Statement I:** Renin converts angiotensinogen into angiotensin I
- Statement II:** An increase in blood flow to the atria of the heart can cause the release of Atrial Natriuretic Factor (ANF).

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.
39. What is the process of excreting urine known as?
- (1) Diuresis
 - (2) Polyuria
 - (3) Defaecation
 - (4) Micturition

40. Which of the following statements is **incorrect**?
- (1) 1200 mOsm/L is the osmolarity of inner medulla of the kidney.
 - (2) Average weight of kidney is about 120-170 gm.
 - (3) An adult human excretes, on an average, 1 to 1.5 litres of urine per day.
 - (4) Vasa recta is absent or highly reduced in juxta medullary nephrons.

41. The fluid filtered from the glomerular capillaries into the Bowman's capsule is equivalent to:
- (1) plasma without proteins.
 - (2) plasma with proteins
 - (3) blood without formed elements
 - (4) serum

42. Which of the following is an **incorrect** pair regarding urine analysis and its diagnostic significance?
- (1) Glycosuria - Presence of glucose in urine
 - (2) Diuresis - Increase urine production
 - (3) Uremia - Accumulation of urea in blood
 - (4) Ketonuria - Indicative of diabetes insipidus

43. Match **List-I** with **List-II**.

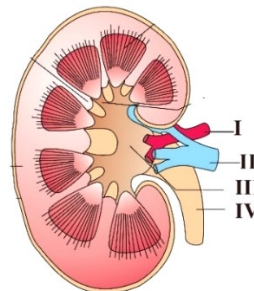
List-I		List-II	
(A)	Ammonotelic	(I)	Renal tubule
(B)	Uricotelic	(II)	Reptiles
(C)	Ureotelic	(III)	Marine fishes
(D)	Bowman's capsule	(IV)	Bony fishes

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
 - (2) A-II, B-I, C-IV, D-III
 - (3) A-III, B-IV, C-II, D-I
 - (4) A-IV, B-II, C-III, D-I
44. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Ascending limb of loop of Henle is impermeable to water and allow transport of electrolyte actively and passively.
Reason R: Dilution of filtrate takes place due to efflux of electrolyte in the medullary fluid.
 In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

45. The diagram given below shows longitudinal section of kidney with labelled **I, II, III, and IV**. Which of the following represents the renal pelvis?



- (1) I
- (2) II
- (3) III
- (4) IV

46. Reabsorption of water from distal parts of tubule is facilitated by hormone _____.

- (1) vasopressin
- (2) ADH
- (3) aldosterone
- (4) both (1) and (2)

47. Read the following statements (**A-E**), and select the **incorrect** statement.

- Excessive loss of body fluid from the body switch off osmoreceptor.
- ADH cause vasodilation.
- ADH increase blood pressure.
- ADH is responsible for decrease in GFR.
- Kidneys are situated between the last thoracic and third lumbar vertebra.

Choose the most appropriate answer from the options given below:

- (1) A, B and D only
- (2) A, B, C and E only
- (3) B, C and D only
- (4) A, B, C, D and E only

48. Which of the following sphincters controls the release of urine?

- (1) Cardiac sphincter
- (2) Pyloric sphincter
- (3) Urethral sphincter
- (4) Oesophageal sphincter

